CDA 106
Numerical Methods for Data Science

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Power Method (Iterative method)

Idea initialize $x_0 \neq 0$, $x_1 = Ax_0$, $x_2 = Ax_1$, $x_3 = Ax_2$ ---

after k iterations x_k & $x_{k+1} = Ax_k$

Then, the quotient $q = \dfrac{x_k^T A x_k}{x_k^T x_k} = \dfrac{x_k^T x_{k+1}}{x_k^T x_k} \xrightarrow{\text{approach}}$ the largest eigenvalue in the absolute sense (dominating eigenvalue)

(magnitude)

Let's consider the example

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}$$

eigen values are 1, & 3

$$x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad x_1 = A x_0 = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$k = 1$

$$q = \frac{x_1^T x_2}{x_1^T x_1} = \frac{37}{13} = 2.846$$

$$x_2 = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \end{bmatrix} \; ; \quad x_3 = \begin{bmatrix} A \end{bmatrix} \begin{bmatrix} 5 \\ 9 \end{bmatrix} = \begin{bmatrix} 14 \\ 27 \end{bmatrix}$$

$k = 2$

$$q = \frac{x_2^T x_3}{x_2^T x_2} = \frac{\begin{bmatrix} 5 & 9 \end{bmatrix} \begin{bmatrix} 14 \\ 27 \end{bmatrix}}{\begin{bmatrix} 5 & 9 \end{bmatrix} \begin{bmatrix} 5 \\ 9 \end{bmatrix}} = \frac{313}{106} \approx 2.95 \xrightarrow{\text{approach}} \lambda = 3$$

Again consider the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

$\lambda = 1, 3$

eigenvector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Let's find dominating eigen value using power method.

Sol

$x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $x_1 = Ax_0 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$x_2 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$, $x_3 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 14 \\ 13 \end{bmatrix}$

$x_4 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 14 \\ 13 \end{bmatrix} = \begin{bmatrix} 41 \\ 40 \end{bmatrix}$

$k = 3$

$q = \dfrac{x_3^T x_4}{x_3^T x_3} = 2.997$

$$q = \frac{\begin{bmatrix} 14 & 13 \end{bmatrix} \begin{bmatrix} 41 \\ 40 \end{bmatrix}}{\begin{bmatrix} 14 & 13 \end{bmatrix} \begin{bmatrix} 14 \\ 13 \end{bmatrix}} = \frac{1094}{365} = \underline{\underline{2.997}}$$

$$\approx \boxed{\lambda = 3}$$

dominant eigen value

as soon as the quotient (q) becomes <u>closer</u> to each other at ~~consec~~ <u>consecutive</u> iterations, we should stop.

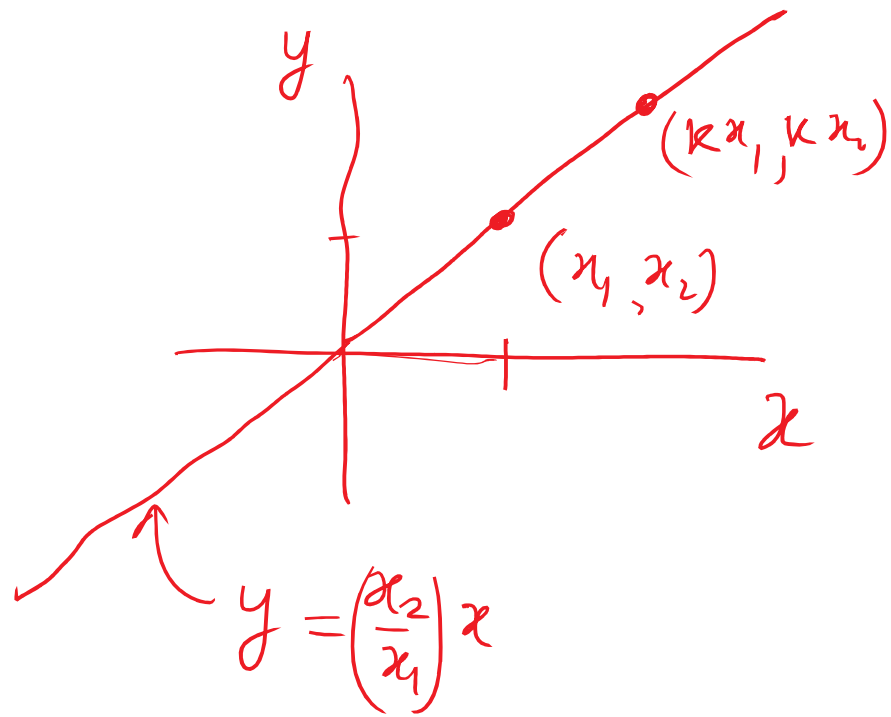
Let us now look at geometrical aspect in the last example.

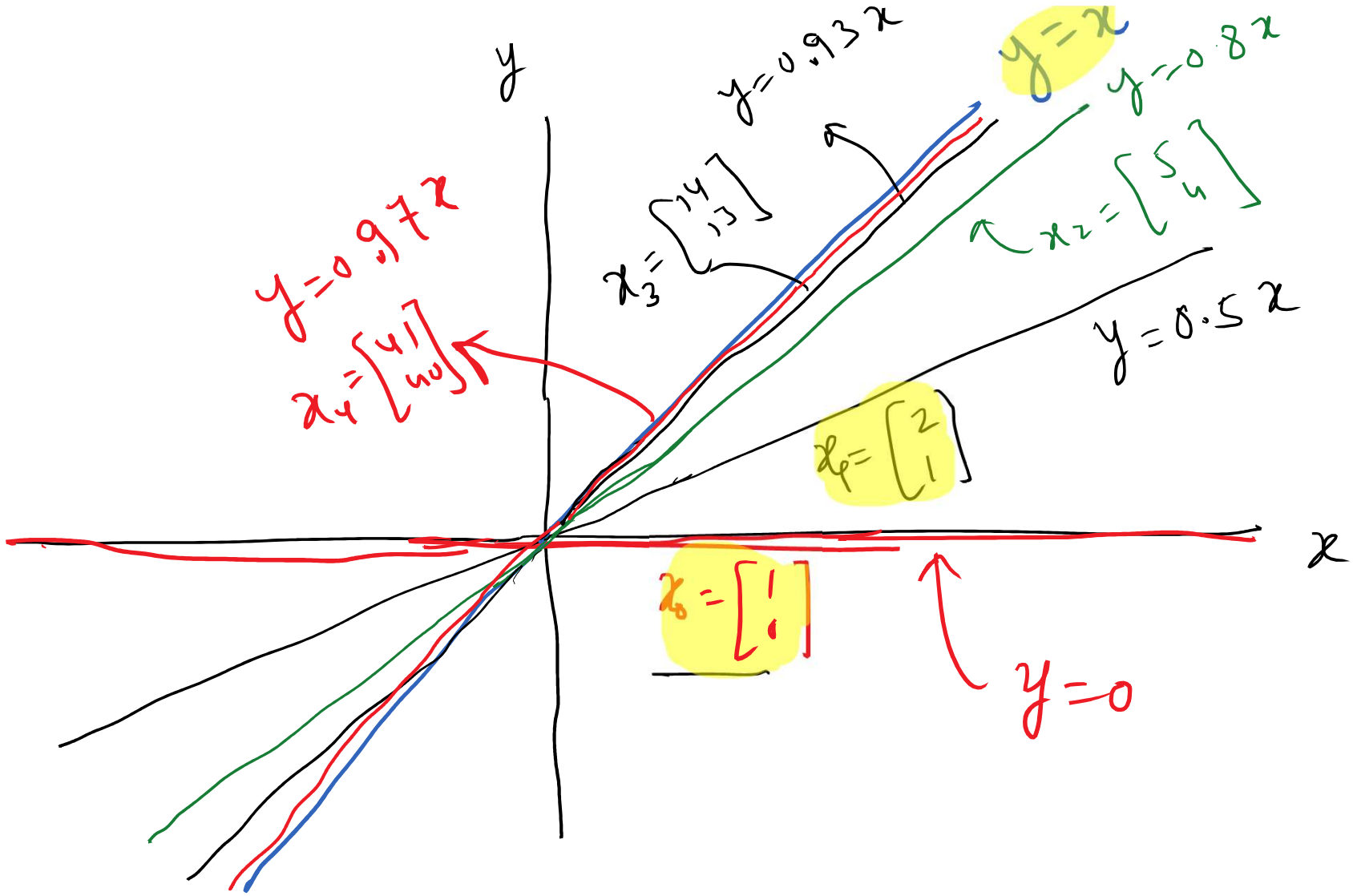
$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \qquad x = x_1 \,,\; y = x_2$$

$$x = k x_1 \,,\; y = k x_2$$

eliminate k

$$y = \left(\frac{x_2}{x_1}\right) x$$




y
x
y = 0.93x
y = x
y = 0.8x
y = 0.5x
y = 0.97x
y = 0
$x_3 = \begin{bmatrix} 14 \\ 13 \end{bmatrix}$
$x_2 = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$
$x_4 = \begin{bmatrix} 41 \\ 40 \end{bmatrix}$
$x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$
$x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Conclusion

(1) Power series method can be used to find dominating eigen value & corresponding eigen vector.

Consider this example for finding eigenvalue & corresponding eigen vector (using power method)

$\rightsquigarrow (\lambda_1 = 1, \lambda_2 = 3)$

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \qquad x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \; ; \quad x_2 = Ax_1 = \begin{bmatrix} 9 \\ 9 \end{bmatrix}$$

$$x_3 = Ax_2 = \begin{bmatrix} 27 \\ 27 \end{bmatrix} = 3^3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad q = \frac{x_2^T x_3}{x_2^T x_2} = \frac{\cancel{2} \times \cancel{9} \times \overset{3}{\cancel{27}}}{\cancel{2} \times \cancel{9}^{\cancel{2}}} = \underline{\underline{3}}$$

eigen vector.

eigenvalue

Power method (Iterative method for finding eigen-values) (Dominating)

Starting vector $x_0 \longrightarrow \underset{= x_1}{Ax_0} \longrightarrow \underset{x_2}{\underset{\shortparallel}{Ax_1}} \longrightarrow \underset{x_3}{\underset{\shortparallel}{Ax_2}} \cdots$

"Arbitrary"

$$\left\{ \begin{array}{l} \longrightarrow \underline{x = f(x)} \\ x_0 = 0.5 \\ x_{k+1} = f(x_k) \end{array} \right.$$

Example

$$A = \begin{bmatrix} 9 & 10 & 8 \\ 10 & 5 & -1 \\ 8 & -1 & 3 \end{bmatrix}$$

Apply Power method to get dominating eigen value

Ans

Assume

$$x_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$X_1 = AX_0 = \begin{bmatrix} 9 & 10 & 8 \\ 10 & 5 & -1 \\ 8 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 27 \\ 14 \\ 10 \end{bmatrix}$$

$$X_2 = \begin{bmatrix} 9 & 10 & 8 \\ 10 & 5 & -1 \\ 8 & -1 & 3 \end{bmatrix} \begin{bmatrix} 27 \\ 14 \\ 10 \end{bmatrix} = \begin{bmatrix} 463 \\ 330 \\ 232 \end{bmatrix}$$

$$X_3 = \begin{bmatrix} 9323 \\ 6048 \\ 4070 \end{bmatrix}; \quad X_4 = \begin{bmatrix} 176947 \\ 119400 \\ 80746 \end{bmatrix}, \quad X_5 = \begin{bmatrix} 3432491 \\ 2285724 \\ 1538414 \end{bmatrix}$$

$$X_6 = \begin{bmatrix} 66056971 \\ 44215116 \\ 29781446 \end{bmatrix} \qquad \underline{X_7} = \begin{bmatrix} 1.27\times10^{9} \\ 8.51\times10^{8} \\ 5.73\times10^{8} \end{bmatrix}$$

$$Y = \underline{A}X_6 = X_7$$

quotient $$q = \frac{X_6^T \underline{A} X_6}{X_6^T X_6} = \frac{X_6^T X_7}{X_6^T X_6} = \frac{1.3896\times10^{17}}{7.206\times10^{15}} \approx 19.28$$

λ

In this process we note, that if the entries of matrix A are Larger (2 are of same sign), then the calculation of X_j's will become difficult. In such cases, one of the sol^n is ~~Scaling~~ "Scaling".

Power Method, Error Bounds

Let $\mathbf{A}$ *be an* $n \times n$ *real symmetric matrix. Let* $\mathbf{x}\ (\neq \mathbf{0})$ *be any real vector with* n *components. Furthermore, let*

$$\mathbf{y} = \mathbf{A}\mathbf{x}, \qquad m_0 = \mathbf{x}^\mathsf{T}\mathbf{x}, \qquad m_1 = \mathbf{x}^\mathsf{T}\mathbf{y}, \qquad m_2 = \mathbf{y}^\mathsf{T}\mathbf{y}.$$

Then the quotient

$$q = \frac{m_1}{m_0} \tag{1}$$

(Rayleigh[10] quotient)

is an approximation for an eigenvalue λ *of* $\mathbf{A}$ (usually that which is greatest in absolute value, but no general statements are possible).

Furthermore, if we set $q = \lambda - \epsilon$, *so that* ϵ *is the error of* q, *then*

$$|\epsilon| \leqq \delta = \sqrt{\frac{m_2}{m_0} - q^2}. \tag{2}$$

Example Find the largest eigen value using Power Method

for $A = \begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix}$ $\lambda = -2, -1, 0$

Ans $x_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ then $X_1 = \begin{bmatrix} A \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \\ 4 \end{bmatrix}$

$X_2 = AX_1 = \begin{bmatrix} 2 \\ 0 \\ -4 \end{bmatrix}$, $X_3 = AX_2 = \begin{bmatrix} 0 \\ -2 \\ 4 \end{bmatrix}$, $X_4 = \begin{bmatrix} -4 \\ 6 \\ -4 \end{bmatrix}$

$X_5 = \begin{bmatrix} 12 \\ -14 \\ 4 \end{bmatrix}$ $X_6 = \begin{bmatrix} -28 \\ 30 \\ -4 \end{bmatrix}$ $X_7 = \begin{bmatrix} 60 \\ -62 \\ 4 \end{bmatrix}$, $X_8 = \begin{bmatrix} -124 \\ 126 \\ -4 \end{bmatrix}$ $X_9 = \begin{bmatrix} 252 \\ -254 \\ 4 \end{bmatrix}$

$= AX_8$

Assume $\underline{Y = AX_8}$ $\quad \underline{X = X_8}$

$$m_1 = X^T Y = \begin{bmatrix} -124 & 126 & -4 \end{bmatrix} \overset{Y}{\begin{bmatrix} 252 \\ -254 \\ 4 \end{bmatrix}} = -63,268$$

$$m_0 = X^T X = \begin{bmatrix} -124 & 126 & -4 \end{bmatrix} \begin{bmatrix} -124 \\ 126 \\ -4 \end{bmatrix} = 31,268$$

quotient $q = \dfrac{m_1}{m_0} = \dfrac{-63,268}{31,268} \approx \underline{-2.0234} \longrightarrow \lambda$ (dominant eigenvalue)

$$m_2 = Y^T Y = \begin{bmatrix} 252 & -254 & 4 \end{bmatrix} \begin{bmatrix} 252 \\ -254 \\ 4 \end{bmatrix} = \underline{128036}.$$

$$|\epsilon| \le \sqrt{\frac{m_2}{m_0} - q^2} = \sqrt{\frac{128036}{31268} - q^2}$$

$$= \underline{0.024}$$

Let us look at another calculation

($\lambda = -20$ as largest eigenvalue)

Exact

$$A = \begin{bmatrix} -15 & 4 & 3 \\ 10 & -12 & 6 \\ 20 & -4 & 2 \end{bmatrix} \quad \text{choose } X_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$X_1 = AX_0 = \begin{bmatrix} -8 \\ 4 \\ 18 \end{bmatrix} \xrightarrow{\text{Scale } X_1} \begin{bmatrix} -4/9 \\ 2/9 \\ 1 \end{bmatrix} \equiv Y_1$$

$$\lambda \approx q = \frac{X_0^T A X_0}{X_0^T X_0} = \frac{\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -8 \\ 4 \\ 18 \end{bmatrix}}{3} = \frac{14}{3} \approx \underline{\underline{4.67}}$$

$$X_2 = A Y_1 = \begin{bmatrix} -15 & 4 & 3 \\ 10 & -12 & 6 \\ 20 & -4 & 2 \end{bmatrix} \begin{bmatrix} -4/9 \\ 2/9 \\ 1 \end{bmatrix} = \begin{bmatrix} 95/9 \\ -10/9 \\ -70/9 \end{bmatrix} = X_2$$

Scale X_2

$$q \cancel{\approx} \frac{X_1^T A X_1}{X_1^T X_1} \approx \frac{\cancel{X_1^T} X_2}{\cancel{X_1^T} Y_1} \approx \cdots$$

$$Y_2 = \begin{bmatrix} 1 \\ -2/19 \\ -14/19 \end{bmatrix}$$

Keep on doing the calculations, $X_7 = \begin{bmatrix} -19.867 \\ 9.707 \\ 19.05 \end{bmatrix}$

$$\frac{Y_6^T X_7}{Y_6^T Y_6} \approx q$$

$$q \cong -19.97$$

Again consider the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ $\lambda = 1, 3$

eigenvectors $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Let's find dominating eigen value using power method.

Sol $x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $x_1 = Ax_0 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} = Y_1$

$$x_2 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.8 \end{bmatrix} \approx Y_2$$

$$X_3 = \underset{A}{\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}}\underset{Y_2}{\begin{bmatrix} 1 \\ 0.8 \end{bmatrix}} = \underset{X_3}{\begin{bmatrix} 2.8 \\ 2.6 \end{bmatrix}}$$

Therefore $q = \dfrac{Y_2^T X_3}{Y_2^T Y_2} = \dfrac{4.88}{1.64} = 2.97$

Important

$$q \approx \frac{Y_k^T X_{k+1}}{Y_k^T Y_k}$$

where $\underline{X_{k+1} = A Y_k}$.

Details of the Power method

$A_{n\times n} \longrightarrow$ eigen values of $A = \{\lambda_1, \lambda_2, \ldots, \lambda_n\}$

and corresponding eigen vectors are $\{v_1, v_2, \ldots, v_n\}$

i.e. $\boxed{A v_i = \lambda_i v_i}$ for $i = 1$ to n

Assumption: λ_1 is dominating eigen value

$$|\lambda_1| > |\lambda_2| > |\lambda_3| \ldots > |\lambda_n|$$

Consider $v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n$, $\alpha_i \in \mathbb{R}$

$$\Rightarrow Av = A(\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n)$$

$$= \alpha_1 Av_1 + \alpha_2 Av_2 + \cdots + \alpha_n Av_n$$

$$Av = \alpha_1 \lambda_1 v_1 + \alpha_2 \lambda_2 v_2 + \cdots + \alpha_n \lambda_n v_n \qquad \left[\because Av_i = \lambda_i v_i \ \forall i\right]$$

$$\Rightarrow A(Av) = A(\alpha_1 \lambda_1 v_1 + \alpha_2 \lambda_2 v_2 + \cdots + \alpha_n \lambda_n v_n)$$

$$A^2 v = \alpha_1 \lambda_1 Av_1 + \alpha_2 \lambda_2 Av_2 + \cdots + \alpha_n \lambda_n Av_n$$

$$\Rightarrow A^2 v = \alpha_1 \lambda_1^2 v_1 + \alpha_2 \lambda_2^2 v_2 + \cdots + \alpha_n \lambda_n^2 v_n$$

Keep repeating this process p times to get

& we shall get

$$A^p v = \alpha_1 \lambda_1^p v_1 + \alpha_2 \lambda_2^p v_2 + \cdots + \alpha_n \lambda_n^p v_n$$

or $$A^p v = \lambda_1^p \left[\alpha_1 v_1 + \alpha_2 \left(\frac{\lambda_2}{\lambda_1}\right)^p v_2 + \cdots + \alpha_n \left(\frac{\lambda_n}{\lambda_1}\right)^p v_n \right]$$

Since $\left|\frac{\lambda_2}{\lambda_1}\right| < 1$, $\left|\frac{\lambda_3}{\lambda_1}\right| < 1$ --- $\left|\frac{\lambda_n}{\lambda_1}\right| < 1$, therefore

we know $\lim_{p \to \infty} x^p = 0$ for $|x| < 1$

Therefore, from above eqn we can say as $p \to \infty$

$$\boxed{A^p v = \lambda_1^p \alpha_1 v_1}$$

If we keep on operating A on any v we shall get to the v_1

$$A^p v = \lambda_1^p \alpha_1 v_1 \quad (1) \quad \Rightarrow \quad A^{p+1} v = \lambda_1^{p+1} \alpha_1 v_1 \quad (2)$$

$$(2) \div (1) \Rightarrow \left(\frac{A^{p+1} v}{A^p v} \right) = \lambda_1 \simeq \underline{\underline{q}}$$

Kreyszig (Power method)

* The power method can also be used to find the smallest eigen value.

Q: How can we do that?

Suppose $AX = \lambda X$, $X \neq 0$ $\quad \lambda \rightarrow$ largest eigen value of A

Operate A^{-1} on both sides $\quad A^{-1}(AX) = A^{-1}(\lambda X)$

$$\Rightarrow (A^{-1}A)X = \lambda A^{-1} X$$

$$\Rightarrow IX = \lambda A^{-1} X$$

$$\Rightarrow \frac{1}{\lambda} X = A^{-1} X \quad \text{or} \quad \boxed{A^{-1} X = \frac{1}{\lambda} X}$$

This means $\frac{1}{\lambda}$ is eigen value of A^{-1} & it will be smallest eigen value

Or in other words if I find the largest eigen value of A^{-1} by power method, it will ~~be~~ give smallest eigenvalue of matrix A by taking it's reciprocal.

✓ → Largest eigen value of A (direct use power method)

✓ → Smallest " λ " (find largest e.v. of A^{-1} by power method) $(1/n)$

Correction

Example

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Find the smallest eigen value of A using Power method

Ans. First find A^{-1} and then find largest e.v. using power method.

We note that $A^{-1} = \frac{1}{4}\begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix} \equiv B$.

starting $X_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ $\quad X_1 = BX_0 = \begin{bmatrix} 3/2 \\ 2 \\ 3/2 \end{bmatrix}$

$X_2 = BX_1 = \begin{bmatrix} 5/2 \\ 7/2 \\ 5/2 \end{bmatrix}$, $\quad X_3 = BX_2 = \begin{bmatrix} 17/4 \\ 6 \\ 17/2 \end{bmatrix}$ $\quad X_4 = \begin{bmatrix} 29/4 \\ 41/4 \\ 29/4 \end{bmatrix}$

$X_5 = \begin{bmatrix} 198/16 \\ 280/16 \\ 198/16 \end{bmatrix}$ $\quad X_6 = \begin{bmatrix} 169/8 \\ 239/8 \\ 169/8 \end{bmatrix} = BX_5$ $\quad \underline{X = X_5}, \underline{Y = X_6}$

$$m_0 = X^T X = X_5^T X_5 = \begin{bmatrix} \frac{198}{16} & \frac{280}{16} & \frac{198}{16} \end{bmatrix} \begin{bmatrix} 198/16 \\ 280/16 \\ 198/16 \end{bmatrix} = \frac{39202}{16^2}$$

$$m_1 = X^T Y = X_5^T X_6 = \begin{bmatrix} \frac{198}{16} & \frac{280}{16} & \frac{198}{16} \end{bmatrix} \begin{bmatrix} 169/8 \\ 239/8 \\ 169/8 \end{bmatrix} = \frac{66922}{64}$$

quotient $q = \frac{m_1}{m_0} = \underline{\underline{1.7071}}$ (Largest e.v. of $B = A^{-1}$) $= 1/\lambda$

or smallest eigen value of A

$$= \frac{1}{1.7071} = \underline{\underline{0.5858}} \quad \underline{\underline{\text{Ans}}}$$

<u>Repeat</u> The largest eigen value of A^{-1} will be λ and thenby λ^{-1} will be smallest eigen value of A.

Repeat Argument

$A \longrightarrow \lambda$ smallest eigen value

$$AX = \lambda X \Rightarrow A^{-1}X = \frac{1}{\lambda} X$$

$A^{-1} \longrightarrow \frac{1}{\lambda}$ eigen value (Largest)

Exercise

Find the largest eigenvalue & corresponding eigen vectors

① $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ Ans $\underline{3.41}$ and $\begin{bmatrix} 0.74 \\ -1 \\ 0.67 \end{bmatrix}$ after $\underline{6 \text{ iterations}}$

② $\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ Ans $\underline{3}$ and $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ $\underline{6 \text{ iterations}}$